# HARSH C HARI

Full-Stack Developer & CSE Student

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## SUMMARY

Full-stack developer who builds working, end-to-end systems rather than isolated tutorials — including a 226-endpoint FastAPI platform and a self-hosted cloud gaming orchestrator. Second-year CSE student with hands-on experience in authentication, RBAC, real-time systems, and local AI/LLM integration. Seeking a Software Engineering / Full-Stack Development internship.

## EDUCATION

**Bachelor of Technology, Computer Science and Engineering** 2025 – 2029 (Expected)

*College of Engineering Chengannur, Kerala, India*

CGPA: 8.885 (through Semester 2) | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL

**Frontend:** React, Tailwind CSS, Radix UI (shadcn-style components), HTML/CSS, Chart.js

**Backend:** FastAPI, Express.js, Node.js, REST APIs, WebSockets, JWT Authentication, RBAC

**Databases:** PostgreSQL, Prisma ORM, SQLAlchemy, Firebase

**Testing:** Vitest, React Testing Library

**Tools & Systems:** Git, GitHub Actions, Linux, Vercel, Tailscale, Sunshine

**AI / LLM Integration:** Ollama, Gemma (local LLM serving), Google Gemini API, Prompt Engineering

## PROJECTS

**Personal Cloud Gaming Orchestrator** | *React · FastAPI · Python · WebSockets · JWT · Vitest · Sunshine · Tailscale*

- Built a self-hosted cloud gaming platform: a Python host agent behind a 9-route FastAPI control plane, paired with a React dashboard for remote session control.
- Implemented JWT-secured WebSockets for live updates, plus GPU/system telemetry monitoring and automatic service recovery.
- Integrated Sunshine and Tailscale for low-latency streaming over an unreliable home connection; covered core flows with Vitest and RTL.

GitHub: github.com/Harsh-C-Hari/personal-cloud-gaming-orchestrator

**Productivity Dashboard — Live PWA** | *React · TypeScript · Tailwind CSS · Vite · FastAPI · SQLAlchemy · PostgreSQL · JWT · Web Push · Google OAuth*

- Engineered a full-stack project and study-management platform — 226 REST endpoints across 40 FastAPI routers — covering tasks, projects, role-based permissions, bug tracking, and a knowledge base.
- Designed a Web Push system (VAPID + service worker) that delivers deadline alerts even with the app closed, backed by a GitHub Actions cron job every 5 minutes and server-side dedup.
- Implemented server-verified Google OAuth and JWT sessions with granular role-based access control, on a Radix UI + Tailwind + React Query frontend.

Live: productivity-dashboard-phi-navy.vercel.app

GitHub: github.com/Harsh-C-Hari/Productivity-Dashboard

**Future of CEC — Campus Platform** | *Node.js · Express · FastAPI · Prisma · PostgreSQL (Supabase) · Ollama · Gemma · Google OAuth · JWT*

- Developed a Student Portal, admin dashboard, and Lift Access Management system integrated into the CEC college website, as part of a 4-person hackathon team ("Minions").
- Designed a role-based request workflow (student request → admin review → approval → automatic expiry) backed by a PostgreSQL/Prisma data layer on Supabase.
- Integrated a local Gemma LLM via Ollama as an AI assistant, routed through Express to a dedicated FastAPI microservice; Google OAuth restricted to institutional email.

GitHub: github.com/Harsh-C-Hari/Minions--Future_of_CEC

**Smart Farming Dashboard — Hackathon Build** | *FastAPI · Python · Chart.js · Firebase · Gemini API*

- Built a dashboard visualizing simulated soil and weather sensor trends with Chart.js, backed by a FastAPI service and Firebase.
- Integrated Google's Gemini API to analyze sensor trends and generate crop-care and next-action recommendations.
- Delivered a working prototype end-to-end within a 24-hour hackathon as part of a 3-person team.

GitHub: github.com/Harsh-C-Hari/build-with-ai-chn-kokachiz

## CERTIFICATIONS

Ethical Hacking (Elite) — NPTEL (SWAYAM), IIT Kharagpur